



Resources to assist in your RISC-V Journey

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What is RISC-V?

RISC-V (pronounced *risk-five*) is:

- An **open standard instruction set architecture (ISA)**
- Based on **reduced instruction set computing (RISC)** principles
- Originally created at **UC Berkeley in 2010**
- Now managed by **RISC-V International**, a global nonprofit

- ✓ Open-standard & Royalty-free
- ✓ Used in real chips and systems
- ✓ Gaining global support across OSes and industries

Why Learn RISC-V?

RISC-V is shaping the future of computing.

- **Open and accessible to everyone**—designed for transparency and collaboration
- **Modular and extensible**—customize it for research, innovation, and real-world solutions
- **Skills in demand**—used across startups, universities, and major tech companies
- **Expanding ecosystem**—hardware, software, tools, and talent all growing fast

Built for the Next 50 Years

- Designed to grow with future tech—from wearables to supercomputers
- Developed through **global collaboration** across 70+ countries
- Backed by a **collaborative & welcoming community** of engineers, researchers, and students
- Ideal for **learning how computers really work** and for building what's next



Training and Resources

Trained over 36,000 people

10 Courses

- Porting Software to RISC-V (FREE) (NEW)
- Open Source RT-Thread RTOS on RISC-V (FREE)
- Introduction to RISC-V (FREE)
- RISC-V Fundamentals (\$99)
- Building a RISC-V CPU Core (FREE)
- And more! <https://riscv.org/community/training/>



1 Certification



RISC-V Community Challenge

Community Challenge: A University Partnership for the Global RISC-V Community

Our first Community Challenge: Build a RISC-V Microcontroller from the Ground Up with HaDes-V

Over a few months, using a modular, step-by-step approach, participants designed a pipelined 32-bit RISC-V microcontroller, validated each stage with golden reference modules, and gained practical experience with SystemVerilog, verification, and debugging tools.

Check out our first Community Challenge here 





RISC-V Developer Workshops

Developer Workshops : Hands-On Workshops

- For developers currently working on RISC-V or interested in increasing their knowledge
- Allows you to move beyond theoretical knowledge to direct application
- Experiment with tools
- Taught by industry experts
- In-person at RISC-V Summit or watch the recordings and follow along with handouts

Check out the recordings on YouTube:

youtube.com/@RISCVInternational



RISC-V Builders

What are people actually building with RISC-V?

RISC-V Builders is a YouTube Live series where the people behind a project get together to talk about what they built, how they built it, and what they learned along the way.

In the first episode, a professor and his student talked about how they turned HaDes-V, a teaching project, into Tartarus, a RISC-V system capable of running DOOM.

Check out the first episode here 





Featured Work

Curious about what others are building on RISC-V?

Featured Work is a blog series that showcases projects from PhD students, professors, and universities around the world:

- Real-world applications, hardware security, ...
- Research breakthroughs
- Innovations happening right now with RISC-V

Whether you're deep into your PhD or just getting started, it's a great way to get inspired, share your own work, and see how the global community is shaping the future of open computing.

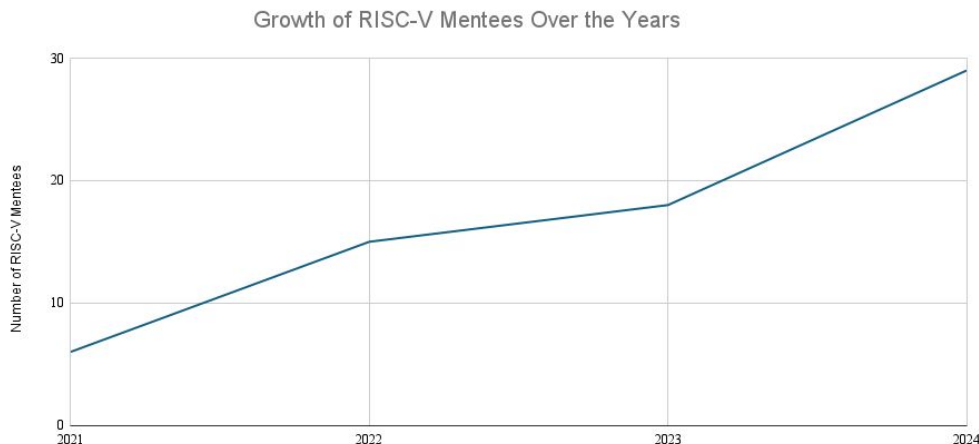
Check out the latest projects here 





Paid Mentorships

Mentorships: Gain Real-World Work Experience through paid Internships



- 2021: Launch of Paid-Mentorship Program
- 2022: Expansion of the program with the addition of Unpaid Mentorships
- 2025: 101 mentees in total since the program launched 3 years ago



Job Board

Build Your Career With The Open Standard ISA That Is Shaping Tomorrow

- RISC-V is open, accessible, and growing fast. It's a great place to start your career
- RISC-V is powering innovation across industries – from startups to space tech
- The next wave of innovation is happening now and you can help drive it. Whether you're into hardware, software, AI, or embedded systems – there's a place for you here



Learn RISC-V Updates

Monthly Educational Newsletter – 30K subscribers

- Scholarship and Paid-Mentorship Opportunities
- Free Courses, New Courses and Certification(s)
- Events: Hackathons, Career Fairs, Contests
- Highlight projects from/to the community
- Special Discounts
- <https://riscv.org/#newsletter-signup>

Subscribe



Hi Community Member,

Explore paid mentorships, our new free course, and join us at RISC-V Summit Europe for a special bonus. Plus, a recap of the latest RISC-V community projects and a limited-time 40% discount on RISC-V learning!

Let's go! Scroll down 📄

Become a Mentee: Paid Mentorship Opportunities for Summer 2025



Job board: Paid Mentorships - Summer

 RISC-V as a First-Class Citizen in Kernel CI - Part II Apply	 and other participants of RISC-V Performance Modeling SL Create a Workload Analysis Flow Using RISC-V Olympia Performance Model Apply	 Alternate Numerical Notations for RISC-V Apply	 Hardware Abstraction Layer Transitional Library for Software to Hardware Ports For RISC-V - Part II Apply
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Summer Term: June 1 - August 31, 2025

Applications Deadline: May 16

Check out the [paid mentorship opportunities](#) with RISC-V Member Organizations and gain real-world work experience. Submit your application by May 16.

Apply Now!

Hackathon Online: Are you ready? [Free Event]



For the first time, the [RISC-V Hackathon](#) is online — so anyone, anywhere can join! But if you prefer in-person collaboration, volunteers are hosting local hubs where you can hack together.

➡ Moving forward, all conversations will happen in the [Slack channel](#)!

Hackathon Agenda:

- 🔧 April 1-6: Register and receive **early** access to tooling
- 📅 April 10-14: Gain access to tooling, receive mentorship and start hacking!
- 📺 April 14: Submit your solution via video recording and slides
- 🎤 April 16: Selected solutions presented to the community and judges
- 🏆 April 16: Winners Announced Live

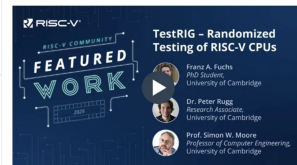
Ready to take on the challenge?

Register Here!



Subscribe

Featured Work: TestRIG – Randomized Testing of RISC-V CPUs



TestRIG is a powerful testing framework for RISC-V CPUs, using model-based verification to compare designs with reference models like Sail RISC-V. With support for RVFI (RISC-V Formal Interface) and DII (Direct Instruction Injection), TestRIG enables rigorous validation and receives contributions from both industry and academia. Explore how it enhances RISC-V reliability in our latest [Featured Work blog post](#).

Read more!



Start Now, Lead the Future

We're here to support your growth from student to innovator:

- Courses and Certifications
- Real-world Projects with Hackathons and Mentorships
- A growing global community
- Opportunities to get noticed through University Demo Day at the RISC-V Summit Europe, Posters, Talks, and "Featured Work"



Join us. Learn RISC-V. Build. Lead

Thank you

www.riscv.org

Who Uses RISC-V?

RISC-V is supported by major companies and used in real products:

- **SiFive, Synopsys, Andes, Alibaba**, and more
- Linux supports it (since kernel 5.17)
- Debian, Fedora, Gentoo, and openSUSE now support RISC-V
- Used in chips, AI engines, embedded systems, and even space tech

Average R&D Spend

10-20%



RISC-V ISA and Ecosystem

Your Innovation

The technical foundation for lasting success

Countries

- Tech sovereignty
- Accelerate local innovation and talent
- Incubate technology ecosystem from research to industry
- Access worldwide market

Multinationals

- Control strategic roadmap
- New opportunities for innovation and influence
- Growth business models
- Avoid vendor lock-in

Researchers

- Collaborative ISA thought leadership
- Enables innovative research
- Access global RISC-V research network
- Tech transfer addressing real world applications

Startups

- Supercharge hardware and software co-design
- Accelerate strategic roadmap in greenfield applications
- Collaboration partners

A shared community that connects us all

Research



Technology



Business

- Companies around the world are pinning their future to RISC-V
- Adoption is growing fast, it's a business and technical priority
- We collaborate and compete
- Let's make the opportunity larger and accelerate success for all



Hackathon Online

RISC-V Hackathon Online

